# HOORVIA - SQL WINDOW FUNCTIONS: PRINTABLE GUIDE

Quick Reference Cheat Sheet · A4 Printable

**Tech & Software · Sheet #1**

| Function | Syntax | What It Does | Example |
|---|---|---|---|
| **ROW_NUMBER** | ROW_NUMBER() OVER (ORDER BY col) | Numbers rows 1, 2, 3... | Top-N per group |
| **RANK** | RANK() OVER (ORDER BY col DESC) | Ranks with gaps for ties | Leaderboards |
| **DENSE_RANK** | DENSE_RANK() OVER (ORDER BY col) | Ranks without gaps | Top-N without gaps |
| **NTILE** | NTILE(4) OVER (ORDER BY col) | Splits rows into N groups | Quartiles, deciles |
| **LAG** | LAG(col, 1) OVER (ORDER BY date) | Previous row's value | Month-over-month change |
| **LEAD** | LEAD(col, 1) OVER (ORDER BY date) | Next row's value | Next order date |
| **SUM (window)** | SUM(col) OVER (PARTITION BY grp) | Running / cumulative total | Running totals |
| **AVG (window)** | AVG(col) OVER (PARTITION BY grp) | Average within a group | Group averages |
| **COUNT (window)** | COUNT(*) OVER (PARTITION BY grp) | Count within a group | Rows per group |
| **FIRST_VALUE** | FIRST_VALUE(col) OVER (ORDER BY date) | First value in a window | First order per customer |
| **LAST_VALUE** | LAST_VALUE(col) OVER (ORDER BY date) | Last value in a window | Latest status |
| **PARTITION BY** | ... OVER (PARTITION BY col) | Resets the window per group | Per-department ranking |